Effects of Dissolved Oxygen Levels on the Greenhouse Gas Emissions and Denitrification Performance of Woodchip Bioreactors Treating Onsite Wastewater

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Abstract

Onsite wastewater treatment systems are used by 25% of the U.S. population, removing nitrogen from household sewage before it re-enters the environment to minimize environmental harm. In particular, denitrification involves the use of woodchip bioreactors, where a variety of microorganisms remove nitrate from wastewater by converting ammonia in incoming wastewater to nitrate then nitrogen gas, leveraging the nitrogen as a nutrient source. However, the use of woodchip bioreactors results in emissions of greenhouse gases, potentially exacerbated by high dissolved oxygen (DO) levels from aeration in nitrification. Thus, this study aimed to test the effects of varying DO concentrations on nitrate removal kinetics and greenhouse gas production in the context of denitrifying woodchip bioreactors. While both low and high DO groups accomplished complete nitrogen removal from wastewater, the low DO levels demonstrated optimal performance. The trends observed in this study between nitrate removal rates and greenhouse gas production at both low and high DO levels were consistent with previous literature: bioreactors operating under low DO levels were associated with higher average nitrate removal rates and reaction rate constants (12.7 g N m^{-3} d^{-1}, k=0.0726) than high DO levels (6.7 g N m^{-3} d^{-1}, k=0.0365). The highest methane concentrations (2026.64 µg C L^{-1}) were observed in the low DO group after complete nitrate removal, while the highest nitrous oxide concentrations (171.91 µg N L^{-1}) were seen in the high DO group and attributed to incomplete denitrification. Since the methane emissions appeared to be preventable by decreasing incubation time, the low DO conditions were shown to be optimal, combining more efficient nitrate removal with less impact on global warming.

Introduction

Importance of nitrogen removal from wastewater

Nitrogenous waste from untreated domestic sewage, as well as from runoff of fertilizers, has been significantly attributed to the development of harmful algal blooms, which utilize this nitrogen as a nutrient source (Diaz-Elsayed et al., 2017; Salo and Salovius-Laurén, 2022). The proliferation of these harmful algal blooms threaten the biodiversity of the waters they inhabit and pose risks to human health (Gobler et al., 2021). Furthermore, throughout the United States, there are an estimated 21.7 million unsewered homes exacerbating this problem (Waugh et al., 2020).

To this end, biological wastewater treatment approaches seek to economically remove nitrogen from wastewater using a wide range of microorganisms which metabolically convert ammonia to nitrogen gas (Diaz-Elsayed et al, 2017; Holmes et al., 2018). Nitrifying bacteria convert the ammonium (NH_4^+) and dissolved oxygen (DO) in wastewater to nitrate (NO_3^-), while denitrifying bacteria reduce the NO_3^- to nitrogen gas (N_2), which is then, in turn, released into the environment (Holmes et al., 2018). For effective nitrogen removal from wastewater, both nitrification and denitrification must occur (Gobler et al., 2021). It is important to remove NO_3^- from water because its ingestion is associated with infant methemoglobenemia, colorectal cancer, thyroid disease, and neural tube defects (Ward et al., 2018).

However, these denitrification reactions may result in the release of nitrous oxide (N_2O; an undesirable greenhouse gas) to the atmosphere as a byproduct of incomplete denitrification, potentially exacerbating the issue of global warming (Davis et al., 2019). These emissions must be further investigated and characterized in order for them to be minimized.

Onsite wastewater treatment systems

Onsite wastewater treatment systems (OWTSs) are used by more than 60 million people and around one-third of new housing developments, providing decentralized sewage treatment for sparsely populated areas without a municipal wastewater treatment plant (Maleki Shahraki et al., 2021; U.S. EPA, 2019). OWTS make up 25% of total wastewater treatment in the U.S. and are often adopted because they are more economically, environmentally, and logistically feasible for certain settings, as centralized facilities require extensive infrastructure and are resource-intensive to maintain (Lee et al., 2021; U.S. EPA, 2019).

Conventional OWTSs are composed of a septic tank and drain field (Figure 1). The septic tank first reduces the concentrations of two important contaminants in influent wastewater: the total suspended solids (by allowing the undissolved particles to settle at the bottom for physical separation) and the carbonaceous biochemical oxygen demand (CBOD—which represents the DO consumed by bacteria and indicates the quantities of aerobic bacteria present) (Diaz-Elsayed et al., 2017). The resulting effluent flows into the drain field, where perforated pipes evenly distribute it throughout the soil for a biological zone (naturally developed over time) containing communities of various microorganisms from wastewater, functioning in parallel, typically to remove 10-40% of total nitrogen (TN) by metabolizing organic matter from soil (U.S. EPA, 2023a; Beal et al., 2005; Cox et al., 2020; Diaz-Elsayed et al., 2017). One of the factors that may contribute to limited nitrogen removal efficiency in this type of OWTS is the depletion of oxygen levels, hindering aerobic microorganisms in nitrification (Cox et al., 2020). Thus, other approaches must be sought to optimize oxygen availability in nitrogen removal.

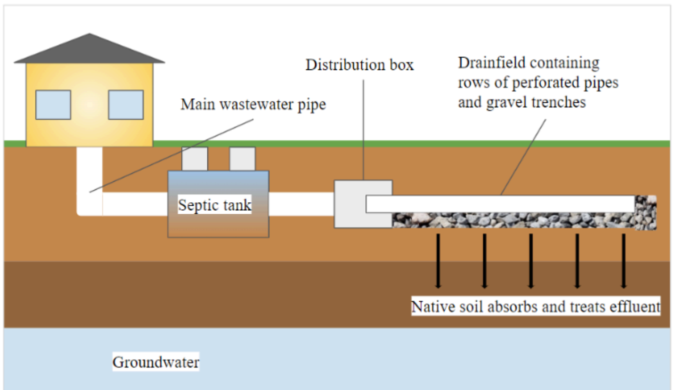


Figure 1. Diagram of conventional septic system. Image generated by student author.

Successes of nitrogen-removing biofilters

Nitrogen-removing biofilters (NRBs) are sometimes used in lieu of the aforementioned drain fields within OWTS, providing improved nitrogen removal at 80-98% of TN as opposed to the 10-40% achieved by conventional OWTSs (Chen et al., 2022; Diaz-Elsayed et al., 2017). NRBs can also be economically connected and integrated with septic tanks, typically serving as in-ground OWTSs in which nitrifying microbes in an aerobic sandy layer and denitrifying microbes in a water-saturated lignocellulose-based layer (made up of woodchips or sawdust, which administer a slow release of carbon to be metabolized by the microbes) transform the NH_4^+, NO_3^-, and nitrite (NO_2^-) in influent wastewater into N_2 (Gobler et al., 2021; Waugh et al., 2020; Li et al., 2016; Maleki Shahraki et al., 2021). In contrast to drain fields (which offer aerobic conditions which are conducive to nitrification but severely limit denitrification), NRBs offer both aerobic and anaerobic conditions, and have thus been attributed to higher TN removal.

Denitrifying microbes, typically heterotrophic bacteria abundant in the natural environment (such as *Bacillus*, *Corynebacterium*, and *Pseudomonas*), enter the NRBs through influent wastewater then colonize the woodchip media in the respective reactors (acting in lieu of the soil in conventional OWTSs, for example) (Mardani et al., 2020). The woodchips elicit the growth of denitrifying microbes by serving as a carbon source and electron donor for their reactions to take place (Mardani et al., 2020). However, not much is known about the release of carbon by the woodchips and its regulating factors, as they are difficult to measure.

One particular implementation of NRBs (involving woodchips) is known as the FlexTreat® biofilter and has achieved TN removal of up to 98% (Figure 2) (Chen et al., 2022). It consists of a pump tank with an air pump (to introduce oxygen to influent wastewater), nitrification columns packed with sand, and denitrification columns containing woodchips, where effluent is recycled back to the pump tank or exits the system (Figure 2). In the pump tank, septic tank effluent is mixed with the recycled denitrification effluent and continuously fed to the surface of nitrification columns. Next, nitrified effluent is fed to the denitrification column from the bottom, flowing upwards and fully saturating the woodchips (which are held down by a weighted net). The denitrifying layer often contains woodchips, which are associated with high NO_3^- removal rates of 80-100% and considerable longevity, with operational time periods of more than 9 years despite little maintenance (Lopez-Ponnada et al., 2017; Moorman et al., 2010). However, while NRBs are highly effective at nitrogen removal from the wastewater itself, they still pose an environmental risk due to the products released—greenhouse gases (Davis et al., 2019).

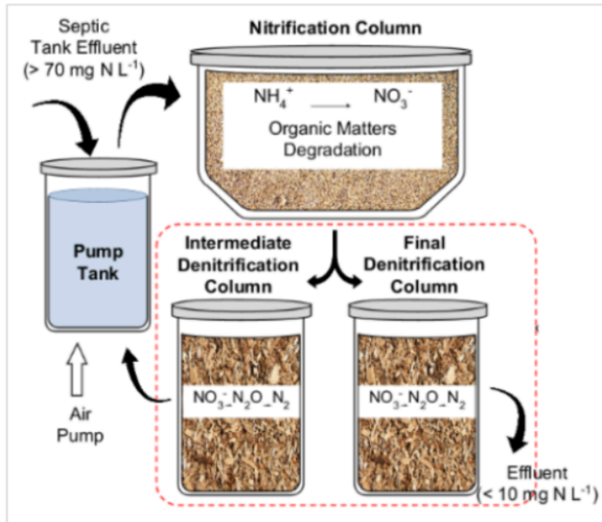


Figure 2. Diagram of FlexTreat® biofilter (Chen et al., 2023).

Greenhouse gas emissions of woodchip bioreactors in NRBs

Woodchip bioreactors are accompanied by environmental concerns due to emission of N_2O, a greenhouse gas 298 times more potent than carbon dioxide (CO_2) in warming the atmosphere and accounts for up to 6% of global warming (Davis et al., 2019; U.S. Audet et al., 2021; EPA, 2023b; Henry et al., 2006). For example, in an NRB achieving TN removal of 80-98%, more than 88% of N_2O production was attributed to the denitrifying woodchip columns (Chen et al., 2022). This is because denitrification takes place in the woodchip columns and N_2O is produced as a result of incomplete denitrification, which happens when NO_3^- is converted to NO_2^-, NO, then N_2O and not N_2 (Davis et al., 2019; Li et al., 2022). This process may remain incomplete due to the inhibition of N_2O reductase, the enzyme converting N_2O to N_2. As such, in a study investigating the impacts of different hydraulic residence times (HRT—average amount of time it takes for a given amount of wastewater to pass through a bioreactor) on greenhouse gas production, N_2O was produced at all three time durations (Davis et al., 2019). In another study investigating the impacts of different HRTs on the N_2O production (of full-scale denitrifying woodchip bioreactors used for agricultural drainage water over the course of several years), N_2O production varied among the different bioreactors (Audet et al., 2021).

Another greenhouse gas emitted by denitrifying woodchip bioreactors, in addition to N_2O, is methane (CH_4)—25 times more potent than CO_2 in warming the atmosphere (Davis et al., 2019; U.S. EPA, 2023b). In a denitrifying woodchip bioreactor, anaerobic conditions were associated with more complete denitrification, reducing N_2O production; however, they also stimulated the production of CH_4 (Elgood et al., 2010). This is due to methanogenesis, a form of anaerobic respiration done by microorganisms (in the domain archea) whose growth is inhibited by oxygen. In methanogenesis, carbon from woodchips is used as the terminal electron acceptor to produce CH_4 and CO_2 (Elgood et al., 2010). Respectively, N_2O and CH_4 are the second and third largest contributors to global radiative forcing, a measurement representing the difference in energy entering and exiting the earth serving as an indicator of global warming (because more energy entering the earth results in more heat) (Davis et al., 2019).

However, these emissions have not been well-characterized: so far, most research has investigated greenhouse gas production in systems using synthetic wastewater (rather than nitrified septic tank effluent) or observed pilot-scale systems whose environments are less controlled due to variation in influent composition and temperatures (Chen et al., 2022; Davis et al., 2019; Robertson, 2010; Elgood et al., 2010). Furthermore, existing research only quantifies the dissolved (and not gaseous phase) greenhouse gases produced due to the absence of a closed system to capture all emissions, which results in the underestimation of total greenhouse gas emissions (Elgood et al., 2010; Davis et al., 2019).

Conditions surrounding DO concentrations may be deterministic of greenhouse gas release

Since oxygen availability influences the organisms which are metabolizing the nitrogenous compounds, the role of DO within these systems may be more consequential than it would seem. Accordingly, in the process of nitrification, DO levels above 2 mg L^{-1} should be maintained for efficient operation (Maleki Shahraki et al., 2021). However, this excessive aeration in the aerobic nitrification chamber (prior to denitrification) may lead to undesirable elevated DO levels in the denitrification chamber (Maleki Shahraki et al., 2021). Higher DO levels are preferred for nitrification, while lower DO levels appear advantageous for denitrification.

Lower DO levels are preferred in denitrification because many denitrifying microorganisms use NO_3^- as their terminal electron acceptor only when oxygen is limited (Maleki Shahraki et al., 2021). This heightened sensitivity to oxygen leads to a detrimental imbalance in the quantities of different enzymes at high DO levels, potentially causing incomplete denitrification where N_2O is released (Henry et al., 2006; Nordström et al., 2021). For instance, a survey of municipal wastewater treatment plants across the United States observed a positive relationship between N_2O emission and DO levels (Ahn et al., 2010).

Additionally, elevated DO levels (3.3-5.8 mg L^{-1} as opposed to 0 mg L^{-1}) afflict the activities of N_2O reductase (the enzyme converting N_2O to N_2 in woodchip bioreactors), since the key denitrification gene nosZ is sensitive to oxygen and denitrification is essentially anaerobic respiration (Chen et al., 2022; Ahn et al., 2010; Spiro, 2012). For example, low NO_3^- removal rates were attributed to high DO levels (3.7-7.3 mg L^{-1}) in a denitrification apparatus using wood-based media and synthetic wastewater (Healy et al., 2006). Moreover, high DO levels were also associated with quicker woodchip degradation, shortening the lifespan of bioreactors (Schaefer et al., 2020). Thus, N_2O release may be modulated by changes in DO levels, making DO a point of interest to ensure that NRBs are not detrimental to the environment in other ways.

Objectives

The goal of this study was to use controlled batch incubation experiments to assess the effects of varying DO concentrations on NO_3^- removal kinetics and greenhouse gas production rates in denitrifying woodchip bioreactors—helping to determine the greenhouse gas emissions of the bioreactors.

Methods

Overview

To simulate closed denitrification columns in NRBs, batch incubation experiments consisting of bottles filled with nitrified effluent (functioning as the wastewater component of an NRB) and a mix of woodchips (functioning as the denitrifying component of an NRB) were set up (Figure 3). Experimental groups were maintained under varying controlled conditions regarding dissolved oxygen (DO) concentrations (Figure 3).

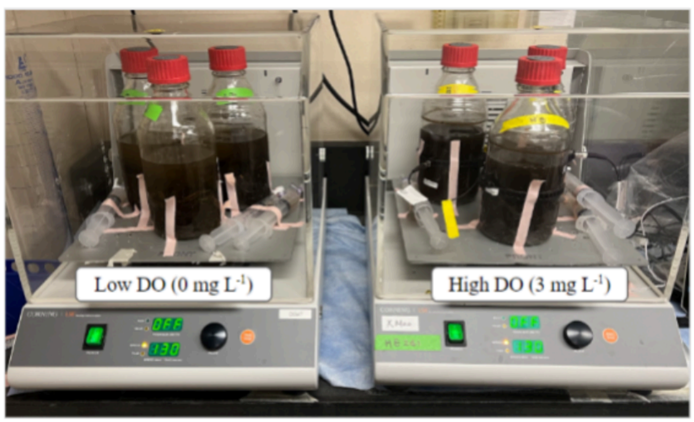


Figure 3. Experimental setup for batch incubation experiment secured onto shakers, with low and high DO groups evaluated in triplicate. Capped with rubber stoppers, 1150 mL bottles contained 700 mL of woodchips fully saturated with nitrified effluent and 450 mL headspace gas. Image generated by student author.

Although they are not typically controlled for, this study regulated DO levels to better assess their effects on the performance of woodchip bioreactors. High DO levels were defined as 3 mg L^{-1}, representing the typical DO of effluent from woodchip bioreactors (Chen et al., 2022). Low DO levels were defined as 0 mg L^{-1}.

Two groups were evaluated in triplicate to discount potential outliers due to inherent error, with each individual trial using a 1150 mL glass bottle capped with an airtight rubber stopper to allow for sample collection using syringes (Figure 3). From this setup, the denitrification kinetics (including TN removal) in the system upon altering initial DO concentration were assessed. Additionally, a second objective of characterizing the release of greenhouse gases (N_2O and CH_4) in the system was investigated to gain novel insight on emissions from denitrifying woodchip bioreactors.

Experimental setup

For each trial, 350 g of woodchips were first inserted into the bottle; then, 400 mL nitrified effluent was poured in, occupying approximately 700 mL of space. This leaves 450 mL of headspace for air. The approximate 7:3 ratio between the volumes of woodchips and headspace gas was derived from that of other woodchip bioreactors (Chen et al., 2022).

Figure 4. Woodchip collection (Chen et al., 2023).

Woodchips were collected from a pilot-scale FlexTreat® biofilter operated for 4 years (Figure 4). During its initial construction, woodchips were obtained from a stockpile located at the Water Research & Innovation Facility in Suffolk County, NY. Woodchip lengths ranged from 0.5-3 cm (Chen et al., 2022). It was necessary to use woodchips acclimated to wastewater treatment (with denitrifying bacteria already present) since they would provide stable operating conditions; truly unused woodchips would not be effective for denitrification (and must be "aged" by running nitrified effluent through them, which would be time-consuming) (Wrightwood et al., 2022).

Nitrified effluent (wastewater that has passed through the nitrification chamber, which carried out complete NH_4^+ removal) was sourced from a pilot-scale FlexTreat® biofilter (depicted in Figure 2) using a valve on the pipe leading from the nitrification column to the denitrification columns. All nitrified effluent in this experiment was obtained at the same time so that its composition would remain consistent across trials and experimental groups.

The DO of the nitrified effluent was initially adjusted down from an initial ambient DO concentration of 5 mg L^{-1} to 0 mg L^{-1} or 3 mg L^{-1}, based on experimental group. This was done by purging the liquid with helium (He) gas (creating a He headspace) then plugging the bottle with a rubber stopper at the desired DO concentration, indicated by a submerged YSI ProSolo Dissolved Oxygen Meter (YSI, USA). However, since DO levels greatly fluctuated during a preliminary experiment, it was necessary to adjust the procedures.

To fix the issue and ensure consistency, the DO concentrations of all bottles were adjusted to 0 mg L^{-1} before either beginning experimentation (for low DO groups) or using a syringe to inject a fixed amount of oxygen gas through the rubber stopper (for high DO groups). Pure oxygen gas was chosen for this purpose because ambient air contains less oxygen per unit volume and may require a larger injection, increasing error by creating leaks in the rubber stoppers. 25 mL of oxygen gas was injected into each bottle for the high DO group at the start of the experiment using a syringe. This number was determined with calculations using a combination of mass balances, Henry's Law, and the Ideal Gas Law—and adjusted and confirmed with preliminary experimentation (Figure S1). Finally, the bottles were put on a shaker at 22°C and 130 rpm (Waugh et al., 2020).

Throughout the experiment, PyroScience Contactless Oxygen Sensor Spots and a FireSting O_2 meter (Aachen, Germany) were used to keep track of DO levels, confirming their consistency (an issue of concern because they rapidly decreased during preliminary experimentation, an effect attributed to CBOD of bacteria). These sensors consisted of a sticker for the inside wall of a glass bottle and a separate device that connects to a laptop, allowing the DO to be monitored remotely without opening the bottle. A 2-point calibration was done with each sensor before every trial, with 100% and 0% oxygen saturation. For calibration, an air pump was first used to raise DO levels until they stabilized at 100% saturation; then, helium gas was added to purge the DO from the liquid until DO levels stabilized at 0% saturation. A calibrated YSI ProSolo Dissolved Oxygen Meter was used to confirm all DO readings during calibration. When DO crossed the membrane into the probe at a rate proportional to its concentration, the luminescence of a dye inside the probe was affected and a photodetector captured the difference created by the change in DO (Yen et al., 2004).

Evaluating denitrification kinetics to assess NO_3^- removal

Since the goal of wastewater treatment is to efficiently eliminate nitrogen from wastewater, denitrification performance was indicated by the concentrations of different nitrogen species over the course of each trial. As the denitrifying microbes removed NO_3^- from the nitrified effluent in the bottle (converting it to NO_2^- N_2O, then N_2), the dissolved NO_3^- concentration decreased and indicated the effectiveness of experimental conditions inside the bioreactor.

Samples from each experimental group were taken over the course of the 80-hour experimental period using 30 mL polypropylene syringes. Every 4-8 hours, a 6 mL liquid sample was collected along with a 24 mL gaseous sample (obtained by inserting the needle and extracting the liquid, tilting the syringe with liquid inside, then extracting headspace air to fill the rest of the syringe). This sample was placed on a shaker (130 rpm and 22°C) for 1 hour. Then, the liquid in the syringe was filtered through a 0.22 μm membrane filter, resulting in ~2 mL of raw samples. After 4 mL of deionized water was added to dilute the sample, the solution was acidified with 10 μL 99% H_2SO_4 for dissolved NH_4^+, NO_2^-, and NO_3^- to be analyzed with a Lachat QC8500 Flow Injection Analysis System (Hach, USA) (Chen et al., 2022). To characterize each analyte, the flow injection analysis system fed the liquid, along with other chemicals, through reaction manifolds then into detectors (which varied depending on the molecules of interest).

The kinetics of denitrification can be explained with the Michaelis-Menten model, where NO_3^- removal follows zero-order kinetics (linearly, where time is the only factor) under high NO_3^- concentrations, and first-order kinetics (where the rate varies based on NO_3^- concentration and can be described by a constant) under low NO_3^- concentrations (Kouanda and Hua, 2021). Although few studies have characterized the NO_3^- removal kinetics of woodchip bioreactors, these kinetics can be used to create parameters optimizing their function.

Quantifying N_2O release of batch incubation experiments

The 24 mL of gas remaining in the syringe was transferred to another syringe then analyzed for dissolved N_2O and CH_4 on a Shimadzu GC-2014 gas chromatograph (Shimadzu, Japan) (Chen et al., 2022).

Every 4-8 hours, 4 mL of gaseous samples were also collected from the headspace of each bottle using a gas-tight syringe, then analyzed for N_2O and CH_4 concentrations on a Shimadzu GC-2014 gas chromatograph (Shimadzu, Japan) (Chen et al., 2022).

By calculating an approximate carbon footprint (CO2e) using the results of this study, the mass of N_2O and CH_4 released was situated in a broader context and compared to other studies and other greenhouse gases (Martindale, 2016). This can be done by normalizing the mass of N_2O and CH_4 released, multiplying them with the Global Warming Potentials (GWPs) of N_2O (298) and CH_4 (25) (Martindale, 2016).

Statistical analysis

Microsoft Excel was used to make unit conversions, calculate averages, and calculate standard deviations for the batch incubation experiments' data. Statistical significance between groups (defined as $p<0.05$) was determined in Microsoft Excel through one-way ANOVA, followed by Tukey's post hoc test (Chen et al., 2022). Pearson correlation coefficients (and coefficients of determination) were also calculated. All graphs were constructed using Microsoft Excel.

Results

The goal of nitrogen-removing biofilters is to efficiently remove nitrogenous compounds from wastewater; however, consequences of doing so have resulted in greenhouse gas release. Thus, this study aims to manipulate oxygen availability to identify whether lower (0 mg L^{-1}) or higher (3 mg L^{-1}) DO concentrations would be optimal in improving nitrogen removal while minimizing greenhouse gas release. Throughout the course of the experiment, all bottles in the high DO group reached DO concentrations of 3 mg L^{-1} and were sustained above 0.5 mg L^{-1} (Figure S2).

Although complete NO_3^- removal was observed for both low and high DO concentrations, low DO concentrations resulted in comparatively higher NO_3^- removal rates

Liquid samples were taken from each bottle at 4-8-hour time intervals over the course of 80 hours. Concentrations of NH_4^+, NO_3^-, and NO_2^- in each sample were analyzed with flow injection analysis, providing information on NO_3^- removal and denitrification kinetics throughout the course of the experiment.

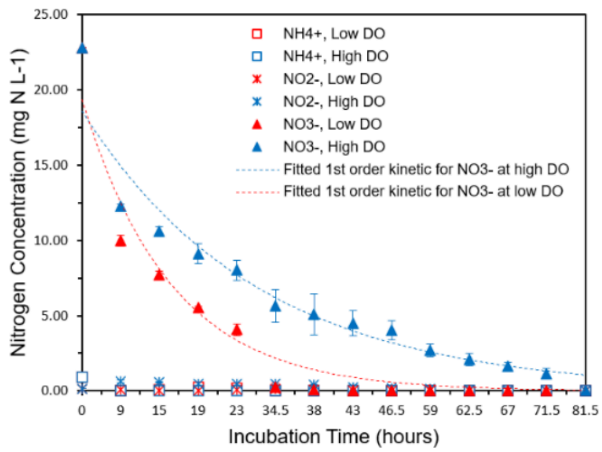
At the start of experimentation, the nitrified effluent (used in both high and low DO concentration groups) had a NO_3^- concentration of 22.76 mg N L^{-1}, NH_4^+ concentration of 0.89 mg N L^{-1} (since it had previously undergone NH_4^+ removal), and NO_2^- concentration of 0.14 mg N L^{-1} (Figure 6). After 80 hours, the concentrations of all nitrogen species (NH_4^+, NO_3^-, NO_2^-) for both groups had decreased to 0 mg N L^{-1}, demonstrating 100% nitrogen removal within the experimental time period (Figure 6).

While both low and high DO groups facilitated complete NO_3^- removal (to 0 mg N L^{-1}), the low DO group did so in nearly half the time of the high DO group—requiring 43 hours compared to 81.5 hours (Figure 6c). The low DO group also had a faster average NO_3^- removal rate, at 12.7 g N m^{-3} d^{-1} compared to the high DO group's 6.7 g N m^{-3} d^{-1}. Thus, low DO levels were more conducive to NO_3^- removal than high DO levels, with respect to initial conditions of the experiment. Meanwhile, NH_4^+ and NO_2^- levels were negligible throughout the course of experimentation—NH_4^+ had already been removed prior (in nitrification), while NO_2^- is an intermediate product produced in small amounts during denitrification.

Additionally, the exponential rates at which the low and high DO groups removed NO_3^- differed. The high DO experimental group had a reaction rate constant (k-value) of 0.0365 for NO_3^-, while the low DO group's was considerably faster at 0.0726 (Figure 6b). The high coefficient of determination ($R^2 >$ 0.98) for each line suggests that these values are representative of the NO_3^- removal rates (Figure 6b). This also indicates that the woodchip bioreactors used in this study obeyed first-order Michaelis Menten kinetics, showing that NO_3^- removal rates depend not only on time but also on the reactant concentration

(Kouanda and Hua, 2021). As the concentration of NO_3^- remaining decreases, the rate at which this NO_3^- is removed also decreases, as seen in the exponential decrease of NO_3^- under both the low and high DO initial conditions (Figures 6a and 6b).

(a)



(b) $A_t = A_0 e^{-kt}$

Group	NO_3^-, Low DO	NO_3^-, High DO
K (hr^{-1})	0.0726	0.0365
A_0 (mg N L^{-1})	21.6325	19.6563
R^2	0.9884	0.9812

(c)

Avg. NO_3^- removal rate (g N m-3 d-1)	6.7	12.7
Time of complete NO_3^- removal	43 hours	81.5 hours

Figure 6. Flow Injection Analysis was used to quantify concentrations of ammonium, nitrite, and nitrate throughout the experiment incubation time of 81.5 hours in order to provide insight on denitrification kinetics. **(a)** Concentrations of nitrogen species over time; error bars represent ±1SD **(b)** Parameters of reaction rate equation for denitrification kinetics **(c)** Average nitrate removal rates prior to complete denitrification and incubation times corresponding to complete nitrate removal. Image generated by student author.

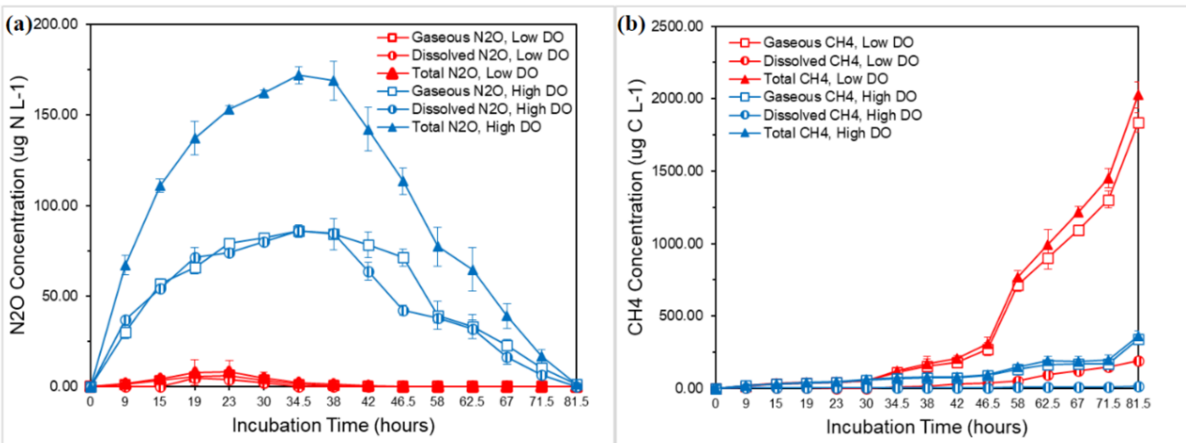
Dependent on initial DO, there is a differential release of greenhouse gases: low DO levels primarily favored CH_4 release, while high DO favored N_2O release

To quantify the greenhouse gas emissions in each bottle, gaseous and dissolved samples were collected using syringes then analyzed using gas chromatography. Gaseous samples were extracted from the headspace, while dissolved samples were obtained by equilibrating nitrified effluent within the bottle with headspace gas.

Overall, lower DO levels corresponded to lower N_2O emissions than high DO groups. This was shown by the maxima of each line graph; the high DO group had a maximum total N_2O concentration of 171.97 µg N L^{-1}, increased from the low DO group's 8.31 µg N L^{-1} ($p < 0.001$) (Figure 7c). Hence, the high DO group produced 20.69 times the maximum total N_2O of its low DO counterpart—sustained at amplified levels for a greater duration of time (Figures 7a and 7c).

Contrarily, lower DO levels were associated with higher CH_4 emissions than higher DO levels were. Although both groups saw an upward trend in CH_4 production as incubation time increased, the low DO group had a maximum CH_4 concentration of 2026.64 µg C L^{-1}, while the high DO group had a maximum CH_4 concentration of 363.67 µg C L^{-1} ($p < 0.001$) (Figure 7c). Moreover, the low DO group produced 5.573 times the maximum total CH_4 of the high DO group, also over a longer time period (Figure 7b and 7c).

Furthermore, N_2O production reached its maximum at earlier incubation times than CH_4 production. N_2O production of low and high DO groups reached peak values at 23 hours and 34.5 hours respectively, while CH_4 production was maximized after more time had elapsed, at 81.5 hours for both time intervals (Figure 7c). Notably, the N_2O graph rose then fell with the increase of incubation time for both low and high DO groups and the CH_4 production of the low DO group increased dramatically upon complete NO_3^- removal at the 43-hour incubation time (Figures 6, 7a, and 7b).



(c)

Group	Max total N_2O concentration (μg N L^{-1}) ($p < 0.001$)		Max total CH_4 concentration (μg C L^{-1}) ($p < 0.001$)	
Low DO	23 hours	8.31	81.5 hours	2026.64
High DO	34.5 hours	171.97	81.5 hours	363.67

Figure 7. Gas Chromatography was used to quantify concentrations of gaseous, dissolved, and total greenhouse gases. **(a)** N2O **(b)** CH4 **(c)** Maximum greenhouse gas concentrations and corresponding incubation times. Error bars represent ±1SD. Two-tailed T-tests were used to evaluate statistical significance. Image generated by student author.

Thus, there appears to be a tradeoff in which both initial DOs result in production of greenhouse gases. As a result, carbon footprints were evaluated by multiplying the masses of N_2O and CH_4 released by their respective global warming potentials (GWPs)—translating impacts of each gas into one common unit (with CO_2 serving as the baseline) (Figure 8). The stacked red columns represent the composition of greenhouse gases released by the low DO group, while the stacked blue columns correspond to the high DO group (Figure 8).

The chart depicts N_2O as the main analyte of interest for the high DO group and CH_4 as the main analyte of interest for the low DO group, since CH_4 and N_2O concentrations were respectively negligible for the high and low DO groups (Figure 8). Moreover, it shows that the carbon footprints of both high and low DO groups varied based on incubation time, rather than DO alone. Generally, lower incubation times (which can also be described as HRT, since incubation times represent how long the wastewater had been inside the bioreactor) resulted in more N_2O production, and higher incubation times resulted in more CH_4 production (Figure 8). Furthermore, low DO levels benefited from shorter HRTs due to less N_2O and therefore less greenhouse gas production in the short-term (and became problematic for global warming if run with too high of an HRT), while high DO groups benefited from longer HRTs (Figure 8). It can also be observed that more N_2O was produced during the process of NO_3^- removal, and the woodchip bioreactors began producing CH_4 after complete NO_3^- removal (Figure 8). Given the time it takes for each initial DO concentration group to fully remove nitrogen from effluent, it appears that low DO levels are still less harmful to the environment.

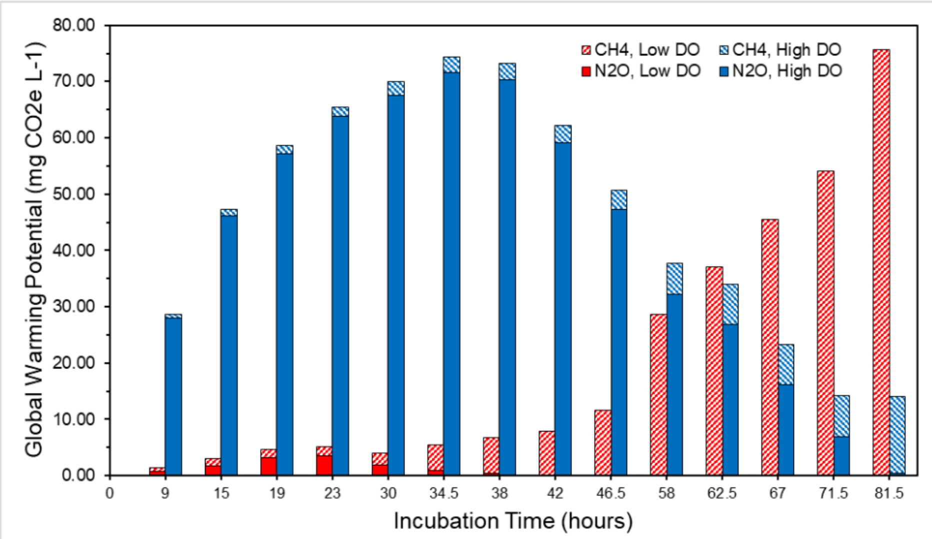


Figure 8. Total (gaseous and dissolved) greenhouse gas concentrations were obtained, stratified (between N2O and CH4), then converted to CO2 equivalents (multiplying by global warming potential, obtaining the carbon footprint or global warming impact) to be depicted in the column chart for low and high DO groups in the experiment (Martindale, 2016). Image generated by student author.

Discussion

The goal of this study was to use controlled batch incubation experiments simulating denitrifying woodchip bioreactors treating nitrified wastewater to assess the effects of varying DO concentrations on denitrification kinetics and greenhouse gas production.

NO_3^- removal was more effective under low DO conditions, rather than high DO conditions

Low DO conditions were more conducive to NO_3^- removal than high DO conditions were; the low DO group exhibited approximately twice the average NO_3^- removal rate at 12.7 g N M^{-3} d^{-1} compared to 6.7 g N M^{-3} d^{-1} (for the high DO group) and approximately twice the reaction rate constant (0.0726 compared to 0.0365) (Figures 6b and 6c). Moreover, the average NO_3^- removal rates of this study generally aligned with those of denitrifying woodchip bioreactors in previous literature; a review presented NO_3^- removal rates ranging from 1.7 to 12.7 g N M^{-3} d^{-1}, and another study observed NO_3^- removal rates from 2.46 to 21.0 g N M^{-3} d^{-1} (Schipper et al., 2010; Kouanda and Hua, 2021). The higher NO_3^- removal rate of 21.0 g N M^{-3} d^{-1} was accomplished by using composted woodchips in a bench-scale bioreactor with an influent NO_3^- concentration of 50 mg N L^{-1}, approximately 2.5 times that of this study

(Kouanda and Hua, 2021). This is consistent with previous literature observing improved nitrogen removal rates with increases in influent nitrogen concentrations (Chen et al., 2022). However, it is difficult to quantitatively compare and contextualize the reaction rate constant observed in this study because few studies have investigated the kinetics of denitrification.

Although the batch incubation experiments performed in this study did not seek to manipulate incubation time, controlled incubation times in bioreactors are commonly referred to as HRT. Therefore, incubation time in this study was analogized to HRT. Since low DO conditions facilitated quicker NO_3^- removal, the low DO group required shorter HRTs (43 hours) than the high DO group (81.5 hours) for complete NO_3^- removal (Figure 6c).

The difference in NO_3^- removal rates between low and high DO groups can be attributed to the inhibitory effect of oxygen on various enzymes in the denitrification process, such as NO_3^- reductase and N_2O reductase—causing the intermediate products (especially N_2O) to accumulate and be released in denitrifying bioreactors (Luo et al., 2016). For example, in a study evaluating the effect of DO on denitrifying bioreactors using biodegradable plastic as a carbon source, low DO levels appeared to support complete denitrification (Zhang and Zhang, 2018). Outside the scope of OWTS, a study evaluating the effect of influent DO on denitrification rates in municipal wastewater treatment plants observed the importance of minimizing influent DO concentrations to maximize denitrification rates (Raboni et al., 2020).

Low DO levels were conducive to high CH_4 production via methanogenesis

In this study, low DO conditions led to quicker NO_3^- removal and consequently, the production of more CH_4 than the high DO group (Figures 6, 7b, 7c, and 8). Accordingly, methanogenesis is performed by microbes who thrive in low DO conditions, where carbon from woodchips is metabolized to CH_4 (Hartfiel et al., 2022). For example, a study saw increased CH_4 production after complete NO_3^- removal in denitrifying woodchip bioreactors (Elgood et al., 2020). Although it modulated NO_3^- removal rates by adjusting temperature, complete NO_3^- consumption was associated with increased CH_4 production during warmer months, in the presence of higher reaction rates (Elgood et al., 2020).

However, methanogenesis is detrimentally affected by the presence of intermediates of the denitrification process (Grießmeier et al., 2017; Klüber and Conrad, 1998; Achtnich et al., 1995; Bollag and Czlonkowski, 1973). In a study observing the effects of NO_3^- concentration on CH_4 production and the activity of methanogens in soil, the presence of NO_3^- (to be metabolized by denitrifiers) hindered production of CH_4, while the absence of NO_3^- spurred CH_4 production (Klüber and Conrad, 1998). This may be due to more efficient use of H_2 (an electron donor) by N_2O-reducing bacteria (along with other microbes) than methanogens; one study observed that H_2 accumulation in soil was blocked by the presence of N_2O (Klüber and Conrad, 1998; Laskowski and Moraghan, 1967). Therefore, denitrifying bacteria outcompete methanogens for the same electron donor, and once intermediate denitrification products (other electron acceptors competing with methanogens) are reduced by H_2, the production of CH_4 increases (Klüber and Conrad, 1998).

While CH_4 was not produced until complete denitrification occurred, the absence of DO increased the functionality of N_2O reductase and complete denitrification, leading to low amounts of N_2O produced in the low DO group (due to the aforementioned activity of N_2O reductase) (Figures 6, 7a, 7c, and 8). Upon complete denitrification, the organic carbon in the woodchips and available H_2 were leveraged for methanogenesis (Hershey et al., 2013).

High DO levels were conducive to high N_2O production due to incomplete denitrification

In this study, N_2O production increased during the process of NO_3^- removal and decreased after complete NO_3^- removal, rising then falling during the incubation time (Figure 7a). N_2O is an intermediate product of denitrification, so its production increases when denitrifying microbes do not transform N_2O into N_2 (Nordström et al., 2021). Moreover, when denitrifiers compete for available organic carbon in municipal wastewater treatment, NO_2^- reductase preferentially receives the electrons over N_2O reductase (Pan et al., 2013). Thus, if a given HRT is not long enough for complete denitrification at a given NO_3^- removal rate, N_2O emissions may result (Audet et al., 2021; Rivas et al., 2020).

The data collected for this experiment indicated that longer HRTs were optimal to minimize the greenhouse gas emissions of the high DO group (Figure 8). In a denitrifying woodchip bioreactor with HRTs of 1-2.6 days, longer HRTs were associated with increased N_2O production (Nordström et al., 2021). Similarly, a study evaluating various HRTs for the FlexTreat® biofilter found that influent DO concentrations were associated with the high DO group in this study (3 mg L^{-1}), and overall TN efficiency was maximized with the longest HRT tested (27.1 hours as opposed to 10 and under) (Chen et al., 2022). Previous literature also suggests that high HRTs are optimal to minimize N_2O emissions: denitrifying woodchip bioreactors facilitating complete NO_3^- removal (at HRTs greater than 60 hours) produced significantly less N_2O than those operating at HRTs of less than 60 hours (Audet et al., 2021). Therefore, DO levels inside denitrifying woodchip bioreactors should be minimized—not only for improved NO_3^- removal, but also for decreased N_2O emissions (Elgood et al., 2020).

Proposal for implementation

Based on the experimental data, low DO operating conditions (0 mg L^{-1}) are generally preferable to high DO conditions (Figure 8). Because variations in lengths of incubation time necessary for each group to reach complete nitrogen removal indicate different ideal HRTs, the optimal HRT for the low DO group in this study was approximately 43 hours, while the optimal HRT for the high DO group was around 82 hours (Figure 6c). However, HRT is not only related to NO_3^- removal—HRT also plays a significant role in dictating greenhouse gas emissions. Shorter HRTs would result in incomplete denitrification and therefore greater N_2O production, while longer HRTs would cause concerns in CH_4 emissions and methanogenesis to arise (Figures 6c and 8).

These optimal HRTs can be calculated for other denitrifying woodchip bioreactors using the first-order kinetic model for NO_3^- removal created in this study, as it can be applied by adjusting the A_0 parameter to correspond to other NRBs' influent NO_3^- concentrations while maintaining the reaction rate constant (k-value). However, this model cannot represent all field-scale systems; for instance, reaction rate would be significantly impacted by temperature (Elgood et al., 2020).

In addition to facilitating more efficient NO_3^- removal, the lower optimal HRTs associated with lower DO concentrations are more economically beneficial, as higher HRTs in bioreactors call for higher construction costs (as they must hold a greater volume of wastewater relative to a given flow rate). Ultimately, both initial DOs were associated with greenhouse gas production—but at the correct HRT, low DO concentrations provided for improved NO_3^- removal and a smaller carbon footprint, thus having a smaller impact on global warming (Figures 6 and 8).

Limitations

Although these incubation experiments seek to simulate real conditions, they do not truly represent the operation of denitrifying woodchip bioreactors in OWTSs. For instance, the pressure inside the bottle was not monitored during experimentation, and was assumed to remain at 1 atm. Since pressure influences the reaction rate (k-value), the reaction rate observed might have been inaccurate (De Persis et al., 2004). Moreover, the composition of nitrified effluent at the incubation time of 0 hours was assumed to be consistent across all bottles and groups, since it was homogenized then poured for experimental setup.

Additionally, the incubation experiments conducted in this study involved homogenizing the nitrified effluent to promote even DO distribution. However, in NRBs, nitrified effluent is not usually homogenized across any given bioreactor; it typically flows upwards, from the bottom to the top of the bioreactor, which would be consequential to the kinetics observed (Chen et al., 2022; Kouanda and Hua, 2021). This is because the highest NO_3^- concentrations in OWTSs would be located at the bottom, decreasing with the increase of distance from the bottom of the reactor. In order to ensure this homogenization, the woodchip/nitrified effluent ratio was decreased from the 7:3 ratio used in the FlexTreat® biofilter (Chen et al., 2022). Due to this difference, the results of this study may overestimate or underestimate greenhouse gas production for FlexTreat® biofilters.

Future Directions

While this study tried to leverage the manipulation of initial DO levels to optimize performance, the impact of available carbon in woodchips should be further investigated: limited amounts of available carbon in woodchip bioreactors operating for long periods of time have been associated with increased N_2O emissions (M. McGuire et al., 2021). As carbon in the woodchips is consumed over time by denitrifying microbes, the overall performance of the bioreactor is reduced—after operating for 1 year, the reactivity of woodchips in a field installation decreased by 50% (Robertson, 2010). Thus, the performance efficiency of woodchip bioreactors may not be sustained as the woodchips inside spend increasing lengths of time treating wastewater, aging within NRBs (Warneke et al, 2011; Davis et al., 2019). Even so, woodchips have overall demonstrated the ability to facilitate stable NO_3^- removal over the course of decades; 7-year and 2-year aged woodchips (which had respectively been used inside functional NRBs for 7 and 2 years) had similar mean NO_3^- removal rates of 12.1±1.7 mg N L^{-1} d^{-1} and 9.1±1.5 mg N L^{-1} d^{-1}, declining approximately 25% over the course of 5 years (Robertson, 2010). Also, a study evaluating a denitrification bed filled with woodchips calculated that there would be enough carbon in the woodchips to sustain up to 39 years of use (Warneke et al., 2011). However, it would be useful to compare the effects associated with the use of woodchips of varying "ages" and observe their characteristics in order to contextualize and explain potential differences in denitrification kinetics and greenhouse gas emissions.

Future research can also seek to normalize the N_2O production of batch incubation experiments so that its impact can be compared with that of other (municipal) wastewater treatment systems utilizing activated sludge in lieu of woodchips, on the basis of our knowledge of the factors which regulate N_2O production. To do this, microbial DNA samples could be isolated from the surface of woodchips in each experimental group, with qPCR being used to measure the abundance of norB and nosZ (gene copies g^{-1})—key functional genes in NO reductase and N_2O reductase, enzymes involved in the formation and dissolution of N_2O during denitrification (Chen et al., 2022; Spiro, 2012).

Additionally, the N_2O production of pilot-scale and full-scale denitrifying woodchip bioreactors can be tested by developing an advanced novel sampling system. This can be done with a tap-like device mounted on the walls of bioreactors to collect gas and liquid samples, then monitor various metrics of wastewater flowing through the system.

Moreover, a mathematical model could be created in the future to predict and optimize the treatment performance of various NRBs. Specifically, a model based on the FlexTreat® biofilter can be developed with data from this study, based on a mass balance evaluation involving the transport and transportation processes of nitrogen species. This model could be used to generate a responsive, automatic adjustment system to control DO levels, balancing nitrification and denitrification efficiency and adjusting for temperature changes throughout seasons. The system would use information from DO sensors in the nitrification and denitrification chambers to control the power of the air pump (used to aerate the wastewater prior to nitrification), and a procedure could be created to adapt it to other NRBs. Although it would be difficult to achieve efficiently and inexpensively, such a system would be beneficial to maximize nitrogen removal while minimizing greenhouse gas emissions.

Furthermore, woodchip bioreactors utilizing low DO levels, as proposed in this study, can implement a system to trap and repurpose CH_4 emissions as natural gas (renewable energy). Previously, anaerobic membrane bioreactors have been used in the context of municipal wastewater treatment to stimulate CH_4 production for this purpose (Song et al., 2018). Thus, methane recovery systems can be applied to woodchip bioreactors, further improving their environmental impact by reducing the carbon footprint and making use of excess energy, outside the context of wastewater treatment.

Conclusion

This study aimed to assess the effects of DO levels on greenhouse gas emissions and denitrification performance in woodchip bioreactors by observing the greenhouse gas evolution and denitrification kinetics over the incubation time period required to fully remove NO_3^- from wastewater under low and high DO conditions. Due to differences in denitrification performance, the low DO group experienced complete NO_3^- removal prior to the high DO group. Therefore, the high DO group experienced a prolonged period of incomplete denitrification (attributed to the inhibitory nature of oxygen on the function of denitrifying microbes), which led to increased N_2O emissions. Although prolonged CH_4 emissions followed complete denitrification for the low DO group, low DO conditions were found to be optimal due to the combination of efficient NO_3^- removal and low greenhouse gas release they maintain at appropriate incubation times.

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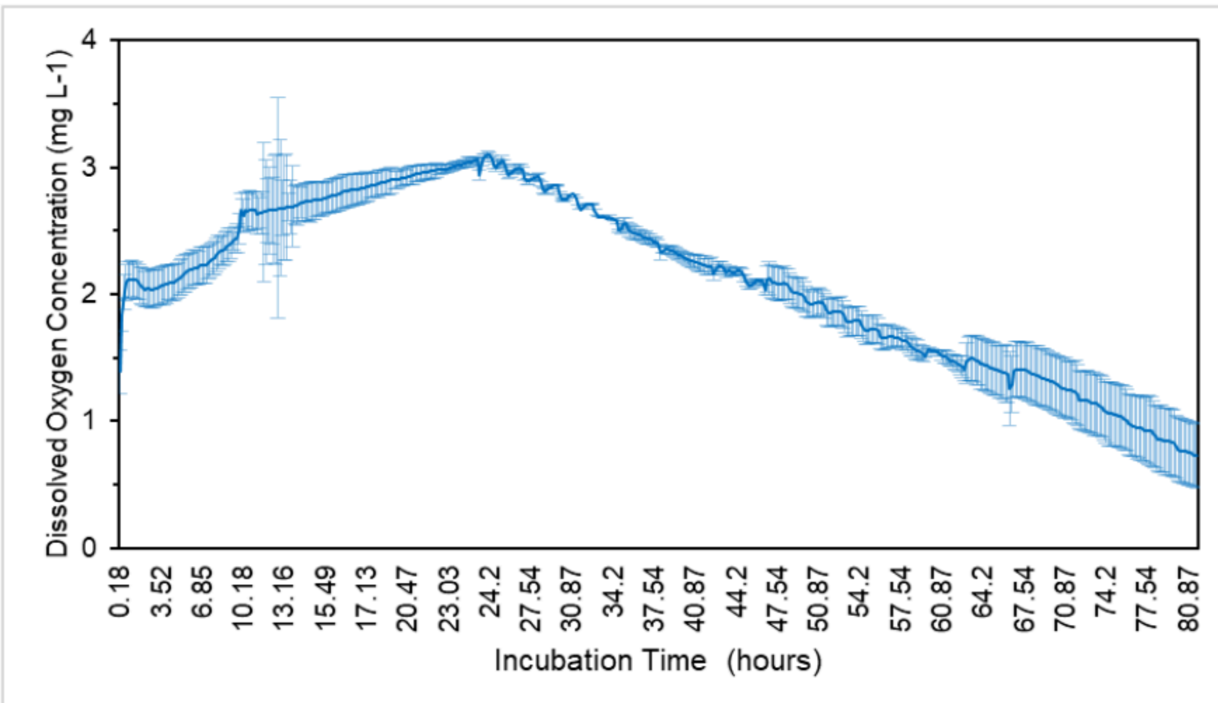
Supplementary Materials

$$M_{O_2} = V_g C_g + V_l C_l \qquad H_{O_2} = 770 = \frac{P_g}{C_l} \qquad PV = nRT$$

Supplementary Figure S1. Equations used to obtain quantity of oxygen for air injection of high DO groups.



Supplementary Figure S2. DO levels throughout the experiment for the high DO group.